Do heat pumps help integrate wind?

Sector coupling and transmission

These solutions are written to be read after the workshop, one subsection per question. Page numbers point to the passage of the paper that answers the question. The paper's figures are not reprinted; the figures here are schematic redrawings, and each caption names the figure of the paper it redraws.

1 What heat pumps cost, and what they do for wind

Part 1 · Reading Ruhnau et al.

1. The research question

Why the questions matter (p. 2). Heat pumps are a main route to decarbonising heat, and wind a main route to decarbonising power; the two meet in the power market. The paper asks whether they help or hinder each other there. If heat pumps are cheap to serve and raise wind's value, electrifying heat also eases the power system's own transition. If not, it is one more load to integrate, on top of wind's variability.

The two opposing forces (p. 2). Heat pumps add demand, more of it in the windy half of the year, which raises prices in the hours wind produces and so wind's value. But the paper holds wind's share of total consumption fixed, heat pumps included (Eqs. (2)–(3)), so more demand means more wind, and the extra wind lowers its value again. Theory cannot say which force is stronger; Fig. 7 separates them.

Beyond the question: which force wins? It is tempting to see the second force as a reaction to the first that can only take back part of the gain. That holds when wind is built on market terms: new wind stops where its value is back down at its cost (p. 12). In the paper the share is set by policy, and then the *shape* of the heat pump load decides. Had it the shape of the rest of demand, adding heat pumps at a fixed share would roughly scale up the whole system, and wind's value would not move. It rises only as far as heat pumps consume in windy hours more than the rest of demand does, and Fig. 5 shows that they do so only weakly.

2. Heat pumps alone

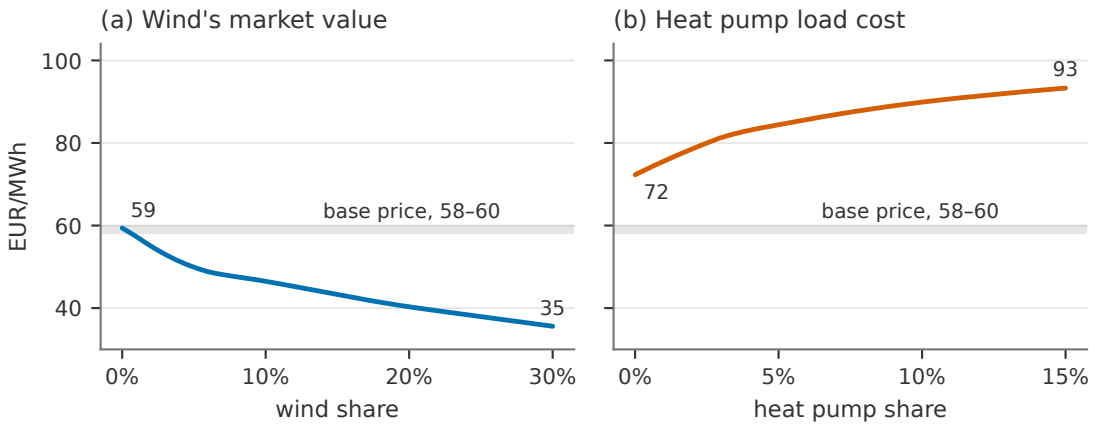


Figure S1: Schematic, after the paper's Fig. 1 (p. 7), with both panels on one price axis and the base price between them: (a) wind's market value against the wind share, without heat pumps; (b) the heat pump load cost against the heat pump share, without wind.

Fig. 1. The two metrics move in mirror image around a base price that stands still. Wind's value falls with its share (by 40% at 30%, p. 5): lecture 3's cannibalisation, as wind's output arrives all at once and depresses the price of its own hours. The load cost rises with the heat pump share (by 29% at 15%) by the same mechanism reversed: inflexible heat pumps all consume in the same cold hours and push up their prices (p. 6). The base price stays put because in a long-run equilibrium it equals the levelised cost of the baseload plant (footnote 10).

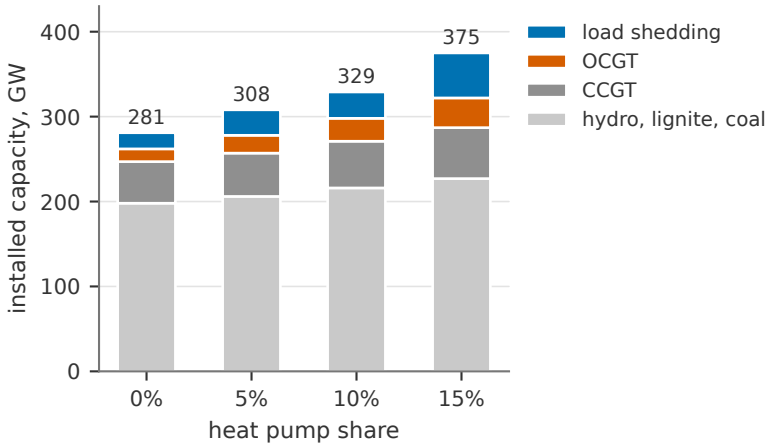


Figure S2: Schematic, after the paper's Fig. 2 (p. 7): installed capacity in the model region by heat pump share, with hydro, lignite and coal as one block.

New capacity (Fig. 2). Capacity grows faster than consumption (by about 30% for a 15% heat pump share, p. 6), and most of what is added serves only the peak: open-cycle gas turbines and load shedding, used a few hours a year. Both follow from the heat pumps being inflexible and volatile. Capacity is built for the peak, and heat pumps raise the peak by more than the average. That extra capacity is the cost the load cost charges them for (p. 6).

Load shedding (Figs. 2–3). Load shedding is demand the model may leave unserved at the value of lost load, which then sets the price. It is the last resort, cheaper than a peaking plant for the few hours a year such a plant would run. At a 15% heat pump share the scarce hours are the heat pumps' own cold hours (Fig. 3, left), so they are the ones paying scarcity prices.

The cost of load shedding barely matters because a higher one makes more peaking plant worth building, which leaves fewer scarcity hours (p. 6). The hint's step: the last peaking plant must cover its fixed cost F per MW from the scarcity hours. With running cost c and scarcity price V, it is built until the N scarcity hours pay for it,

$$N\,(V - c) = F. \tag{1}$$

Double V and N roughly halves, while the rent, F, stays the same (Figure S3). What heat pumps pay through scarcity prices is therefore the fixed cost of the peaking plant built for their peak, whatever the value of lost load. Lecture 6 takes this zero-profit condition up in full (ESM §7).

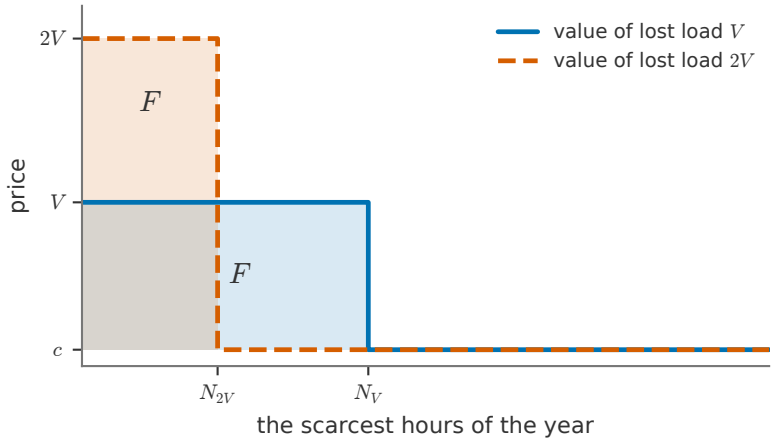


Figure S3: Ours, not the paper's: the scarcest hours of a long-run equilibrium. The last peaking plant is paid by the scarcity rent, an area equal to its fixed cost F, whether the scarcity price is V or $2V$; the doubled price comes with about half the hours.

Beyond the question: the load cost in ESM. ESM §5.3 computes the same statistic for its district-heating heat pump, which has a gas boiler behind it. It pays a little less

than the average price, because the boiler lets it skip the costly hours, while the heat load's own shape would pay well above it (47.0 and 60.6 against 48.8 EUR/MWh). The gap is flexibility, the subject of part 2.

3. Heat pumps and wind together

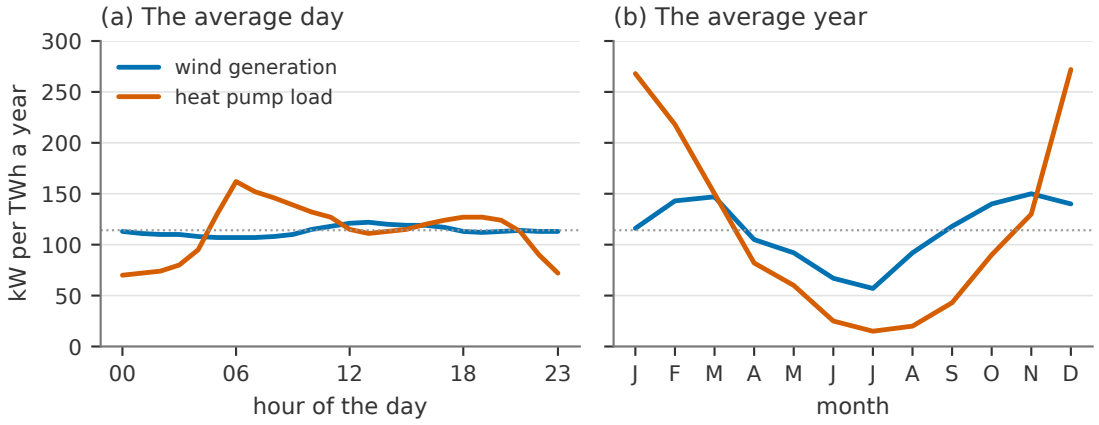


Figure S4: Schematic, after the paper's Fig. 5 (p. 8): wind generation and heat pump load in kW per TWh of annual energy, (a) by hour of the day, (b) by month. The dotted line is a flat profile, $10^9/8760$ kW per TWh.

Fig. 5. The two line up by season, not by hour. Both are higher in winter, the heat pumps much more so, but over the day wind is flat while the heat pumps peak in the morning. A correlation of 0.11 means that the seasonal match the introduction hopes for is there, but within a season the hours hardly match: heat pumps run as hard in a cold, calm week as in a cold, windy one.

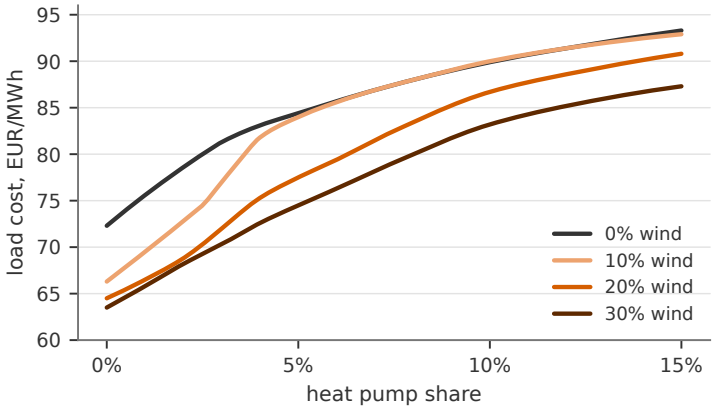


Figure S5: Schematic, after the paper's Fig. 6, right (p. 9): the load cost against the heat pump share, by wind share.

Fig. 6. Wind helps heat pumps far more than heat pumps help wind. Wind lowers the load cost (by almost 10 EUR/MWh at low heat pump shares, p. 6), and less as heat pumps grow and their own cold, calm hours become the scarce ones. Heat pumps barely move wind's value, and not at all at 30% wind.

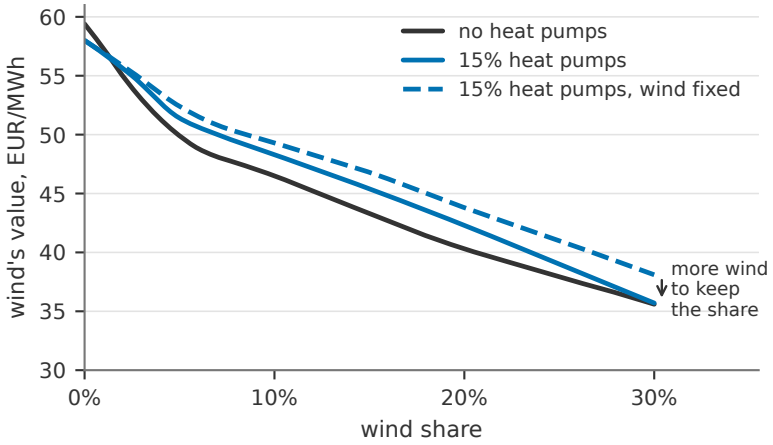


Figure S6: Schematic, after the paper's Fig. 7 (p. 9): wind's market value without heat pumps, with 15% heat pumps at a given wind share, and with 15% heat pumps at the wind capacity of no heat pumps (dashed).

Fig. 7. The dashed line holds wind *capacity* fixed (p. 7), which separates the two forces of question 1. At a fixed capacity only demand grows, more of it in the windy season, and wind's value rises (by 7% at 30% wind). At a fixed share wind's output grows with consumption, and the extra wind cannibalises the gain: partly at 5–20% wind, entirely at 30%. Heat pumps raise wind's value at a fixed capacity, not at a fixed share.

Beyond the question: the value factor, turned around. Divided by the base price, wind's market value at 30% gives a value factor well below one, inside the range Hirth (2013) finds (lecture 3). The load cost gives a "cost factor" that is above one already for the first heat pumps and rises as they grow (about 1.2, then 1.6 at 15%). The start is the telling part: the first heat pumps are too few to move any price, yet cost more than the average MWh, purely because their profile puts their consumption in costly hours. The rise is their own doing, the demand-side mirror of cannibalisation.

Stop point

The load cost is the price weighted by the heat pumps' consumption, the mirror of wind's market value. It rises steeply as heat pumps grow (by some 21 EUR/MWh from nothing to 15%), because a load that follows the heat demand consumes in the scarce hours whose prices pay for the peaking plant built to serve it. Heat pumps raise wind's value at a fixed wind capacity; at a fixed wind share, the extra wind takes the gain back, entirely at 30%.

Stretch question

The load cost as a covariance. With H hours, write the numerator and the denominator of Eq. (5) as averages,

$$LC = \frac{\frac{1}{H}\sum_h \lambda_h l_h}{\frac{1}{H}\sum_h l_h} = \frac{\bar{\lambda}\,\bar{l} + \mathrm{Cov}(\lambda, l)}{\bar{l}} = \bar{\lambda} + \frac{\mathrm{Cov}(\lambda, l)}{\bar{l}},$$

using that the average of a product is the product of the averages plus the covariance. With $\mathrm{Cov}(\lambda, l) = \rho\,\sigma_\lambda \sigma_l$ and the coefficients of variation $CV_\lambda = \sigma_\lambda/\bar{\lambda}$ and $CV_l = \sigma_l/\bar{l}$,

$$\frac{LC}{\bar{\lambda}} \;=\; 1 + \rho\, CV_\lambda\, CV_l.$$

The cost factor exceeds one by how closely the load follows the price (ρ), how volatile the price is, and how peaky the load is. Fig. 4 (Figure S7) shows the last and the first. Low full load hours, the year's consumption divided by its peak, mean a peaky load, a high CV_l and a high cost factor. In 2008 the heat pump load coincided less with the conventional load, which drives prices (footnote 12): a lower ρ, and a low load cost at middling full load hours. A load uncorrelated with the price, or a flat one, costs exactly the base price. For wind, $\rho < 0$ and the same formula gives a value factor below one.

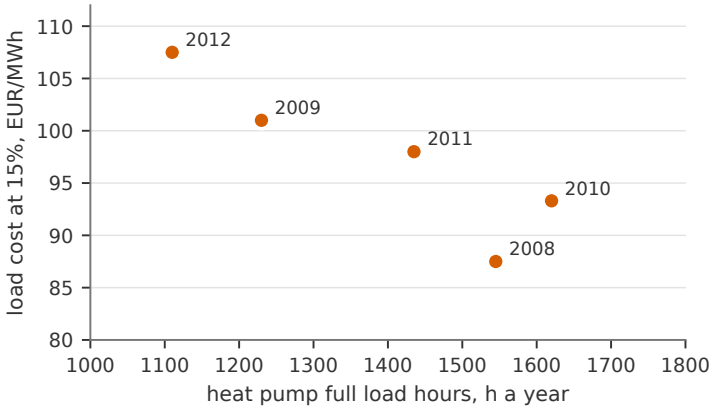


Figure S7: Schematic, after the paper's Fig. 4, right (p. 8): the load cost at a 15% share against the heat pumps' full load hours, by weather year.

2 What takes the cost back, and what it does for wind

Part 2 · Reading Ruhnau et al., continued

1. System-friendly heat pumps

What "system-friendly" means (§3.3). A ground source heat pump with floor heating works across a smaller and steadier temperature difference than an air source pump with radiators, so its coefficient of performance is higher and steadier (footnote 13). For the same heat it draws a *smaller and steadier* load. The paper borrows the term from wind, where turbines with a steadier output are called system-friendly (p. 8).

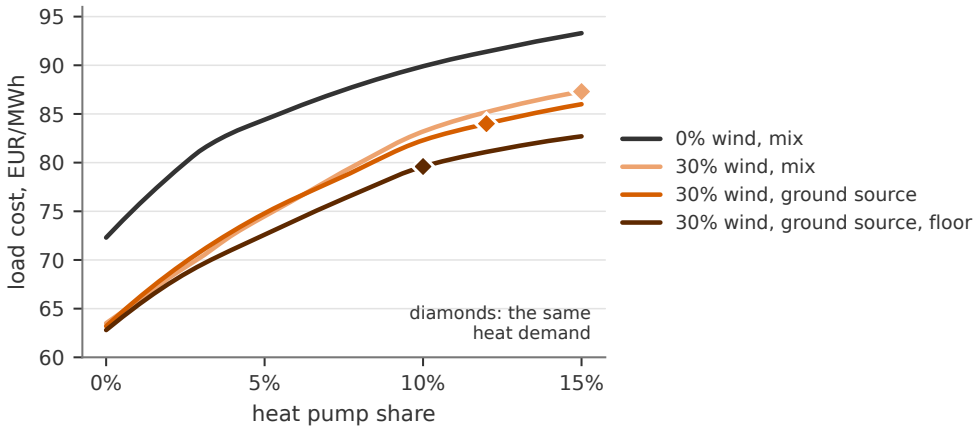


Figure S8: Schematic, after the paper's Fig. 8 (p. 9): the load cost by heat pump technology at 30% wind, with the mix at 0% wind for reference. The diamonds mark the same heat demand: the mix at 15%, ground source at 12%, ground source with floor heating at 10% of consumption.

The load cost (Fig. 8). It falls, for two reasons. A steadier load stays out of the scarcest hours, so it costs less per MWh (compare the curves at the same heat pump share). A smaller load delivers the same heat at a smaller share, further down the curve (compare the diamonds). Together they take a noticeable part off the load cost (about 8 EUR/MWh, p. 8).

Why little for wind. The paper finds no significant effect (p. 8) without saying why; the hint gives the reason. What a steadier load avoids is its peaks, the cold hours with scarcity prices, and those are mostly calm hours in which wind produces little. In the windy hours in which wind earns its value, the heat pump load hardly changes. What makes heat pumps cheaper to serve happens in hours in which wind sells little.

2. Flexible heat pumps

What "flexibilization" means (§3.4). With thermal storage, heat pumps no longer have to follow the heat load: they can respond to prices, consuming in cheap hours and drawing on the store later. But only by a few hours. The store holds two or four hours of peak heat load, and the heat leaks out of it (the stretch box).

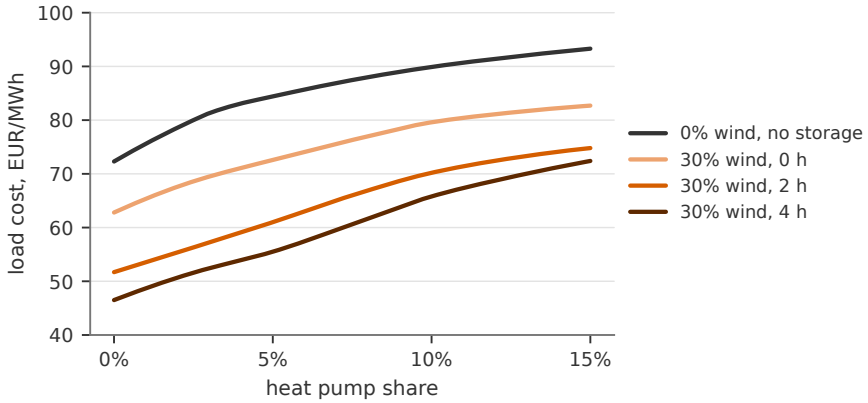


Figure S9: Schematic, after the paper's Fig. 9 (p. 9): the load cost of ground source heat pumps with floor heating at 30% wind, with 0, 2 and 4 hours of thermal storage, and the mix at 0% wind for reference.

The load cost (Fig. 9). Storage lowers it a great deal, by moving consumption out of the costliest hours, but less the more heat pumps there are. Their load then drives prices by season, and a store that holds hours cannot move winter into summer (p. 10).

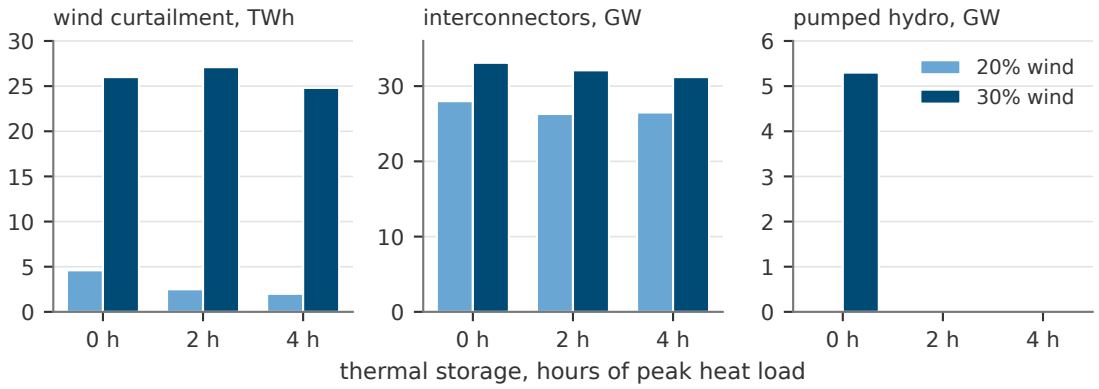


Figure S10: Schematic, after the paper's Fig. 11 (p. 11), split into three panels because the paper plots TWh and GW on one axis: wind curtailment, interconnector capacity and pumped hydro capacity, by hours of thermal storage and wind share, at 15% heat pumps.

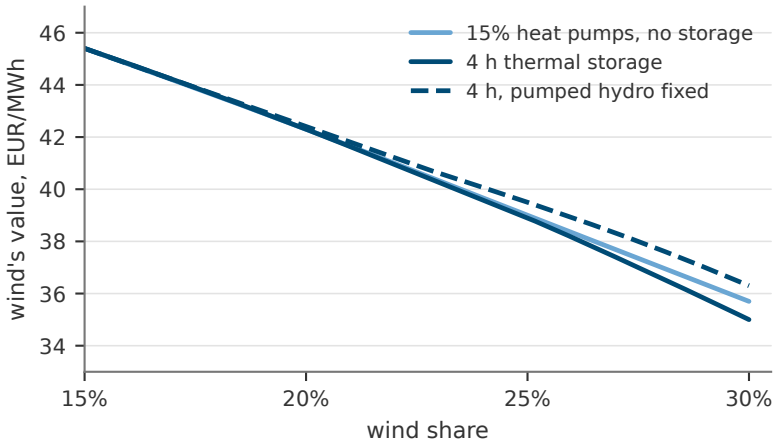


Figure S11: Schematic, after the paper's Figs. 10 and 12 (pp. 10–11), over the wind shares where their lines part: wind's market value at 15% heat pumps without thermal storage, with four hours, and with four hours and pumped hydro held at its level without storage (dashed).

Why little for wind (Figs. 10–12). The heat pumps do move their load towards windy hours: curtailment falls at 20% wind (Fig. 11). Yet wind's value does not rise, and at 30% wind it falls slightly (Fig. 10). Fig. 11 at 30% wind shows why: the storage crowds out interconnectors and all the pumped hydro. Those are paid by the price spreads between zones and between hours, and thermal storage flattens both, so they no longer pay (p. 10). Fig. 12 holds pumped hydro fixed, and then storage does raise wind's value. Thermal storage helps wind on top of flexibility that stays, not in place of flexibility that goes: flexibility options compete rather than add up (p. 12).

Beyond the question: flexibility against flexibility. Lecture 4 met the same substitution. In Zerrahn et al. (2018) a flexible new load, power-to-x, eats the surplus that storage would have stored, and the planner builds a small fraction of the storage. The value of a flexibility option studied on its own overstates what it adds to a system that has the others.

3. What it adds up to

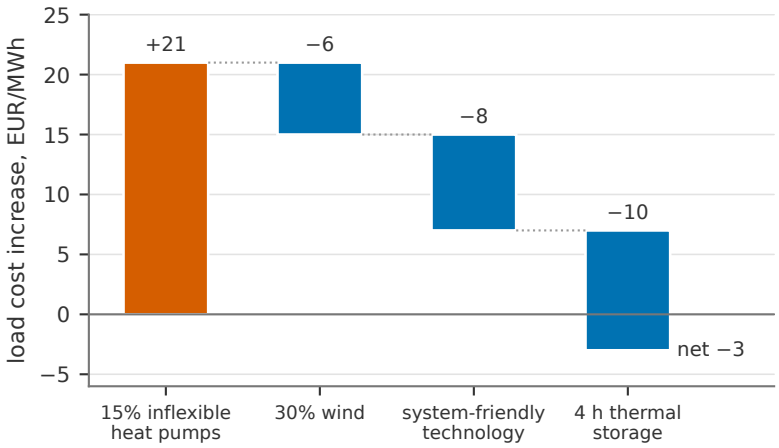


Figure S12: Schematic, after the paper's Fig. 13 (p. 11): the load cost increase of 15% inflexible heat pumps, and what takes it back.

Fig. 13. The waterfall adds up the remedies at a 15% heat pump share: wind, system-friendly technology and four hours of storage. Together they take back more than the whole increase, which is what the authors mean by "overcompensate" (p. 11): heat pumps end up cheaper to serve than the first, inflexible ones were. The rising load cost of part 1 is a property of inflexible air source heat pumps in a system without wind, not of electrified heat.

Summing up. Heat pumps hardly help integrate wind. They raise wind's value only at a fixed wind capacity, and at a fixed share the extra wind takes the gain back; neither better technology nor storage changes that. The help runs the other way: wind lowers the cost of serving heat pumps, and with system-friendly technology and storage more than offsets its increase. For owners to choose those remedies they must be paid for them, and on today's fixed tariffs and standard load profiles they are not (p. 12); hence the paper's call for settlement on wholesale prices through smart meters.

Stop point

System-friendly heat pumps draw a smaller and steadier load, and flexible ones move their load by a few hours. Both lower the load cost, and together with wind they more than offset its increase. Neither does much for wind's value: a steadier load stays out of hours in which wind produces little, and storage crowds out interconnectors and pumped hydro. Wind helps heat pumps more than heat pumps help wind.

Stretch question

How long heat keeps. By Eq. (9), a MWh of heat put in keeps 0.95 of itself on the way in and loses 1% of what is left every hour, so after k hours

$$s_k = 0.95 \times 0.99^k$$

can be taken out: 0.91 after 4 hours, 0.75 after a day and 0.18 after a week. With the same COP in both hours, storing costs the electricity for a MWh of heat at the charging price λ_c and saves the electricity for s_k MWh at the later price λ_d, so it pays only if

$$\lambda_c \;\leq\; s_k\,\lambda_d\,:$$

electricity must be at least 9% cheaper when the storage charges if the heat is held for four hours, 25% cheaper for a day, and 82% for a week. Over a season nothing is left, however large the tank. This storage is a daily store by construction, which is why it helps least where the heat pump load varies most, between seasons.